

# **Amar Telidji University Laghouat**

**Department:** Computer Science

**Level:** 4th Year Engineer

**Course:** Machine learning, Deep learning and Security

**Assignment CH01 Report:** Data Pre-processing and Visualization

**Dataset:** SMS Spam Collection (Kaggle)

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## **1. Introduction:**

This report summarizes the preprocessing steps applied to the SMS Spam Collection dataset and presents visual analyses used to assess dataset readiness for supervised learning. The objective was to produce a clean dataset and a short dashboard indicating whether the data is ready for model training.

## **2. Dataset overview:**

The dataset consists of SMS text messages labeled as either **ham** or **spam**. The Kaggle CSV file contains two main columns:

- **v1:** label
- **v2:** text message

It also includes several unused "**Unnamed**" columns, which are removed during preprocessing.

|   | <b>v1</b> | <b>v2</b>                                         | <b>Unnamed: 2</b> | <b>Unnamed: 3</b> | <b>Unnamed: 4</b> |
|---|-----------|---------------------------------------------------|-------------------|-------------------|-------------------|
| 0 | ham       | Go until jurong point, crazy.. Available only ... | NaN               | NaN               | NaN               |
| 1 | ham       | Ok lar... Joking wif u oni...                     | NaN               | NaN               | NaN               |
| 2 | spam      | Free entry in 2 a wkly comp to win FA Cup fina... | NaN               | NaN               | NaN               |
| 3 | ham       | U dun say so early hor... U c already then say... | NaN               | NaN               | NaN               |
| 4 | ham       | Nah I don't think he goes to usf, he lives aro... | NaN               | NaN               | NaN               |

**HEAD: First 5–10 rows of the raw dataset**

### **3. Preprocessing Steps:**

**Steps performed:**

**1. Column Cleaning**

Removed unused “**Unnamed: X**” columns.

Kept only the two relevant columns: **v1** and **v2**.

**2. Renaming Columns**

- $v1 \rightarrow label$
- $v2 \rightarrow text$

**3. Handling Missing Values**

Dropped any rows with missing text (rare but necessary).

**4. Removing Duplicates**

Removed exact duplicate messages to avoid bias in modeling.

**5. Label Encoding**

Converted categorical labels into numerical form:

- $ham \rightarrow 0$
- $spam \rightarrow 1$

## 6. Feature Engineering

Created a simple numeric feature:

- `text_len` = number of characters in the message

## 7. Outlier Removal

Removed extremely long messages (>1000 characters), which do not represent typical SMS patterns.

## 8. Scaling

Applied standard scaling (**StandardScaler**) to numerical features for consistent range.

|   | label | text                                              | label_num |
|---|-------|---------------------------------------------------|-----------|
| 0 | ham   | Go until jurong point, crazy.. Available only ... | 0         |
| 1 | ham   | Ok lar... Joking wif u oni...                     | 0         |
| 2 | spam  | Free entry in 2 a wkly comp to win FA Cup fina... | 1         |
| 3 | ham   | U dun say so early hor... U c already then say... | 0         |
| 4 | ham   | Nah I don't think he goes to usf, he lives aro... | 0         |

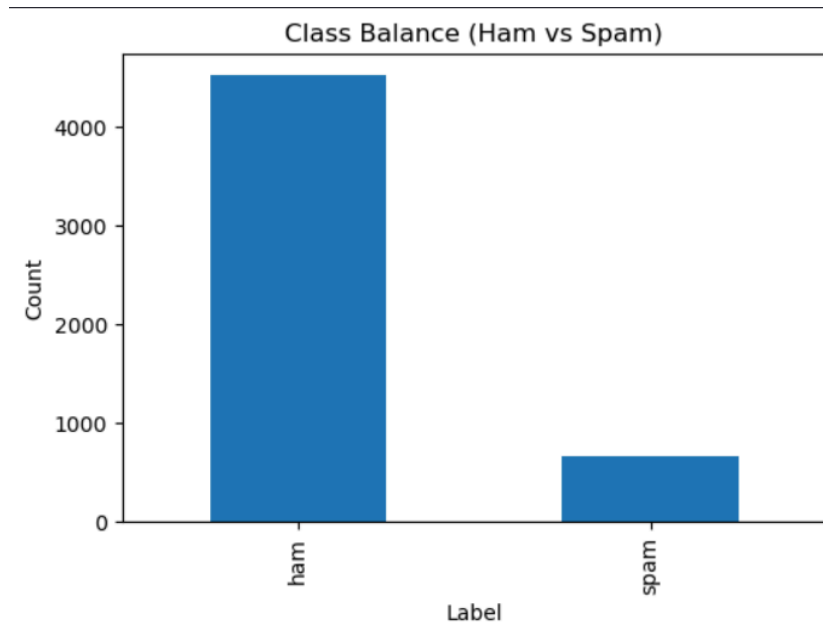
Output showing dataset shape after cleaning

## 4. Visualization Dashboard:

Several visualizations were produced to assess dataset readiness and understand data distribution.

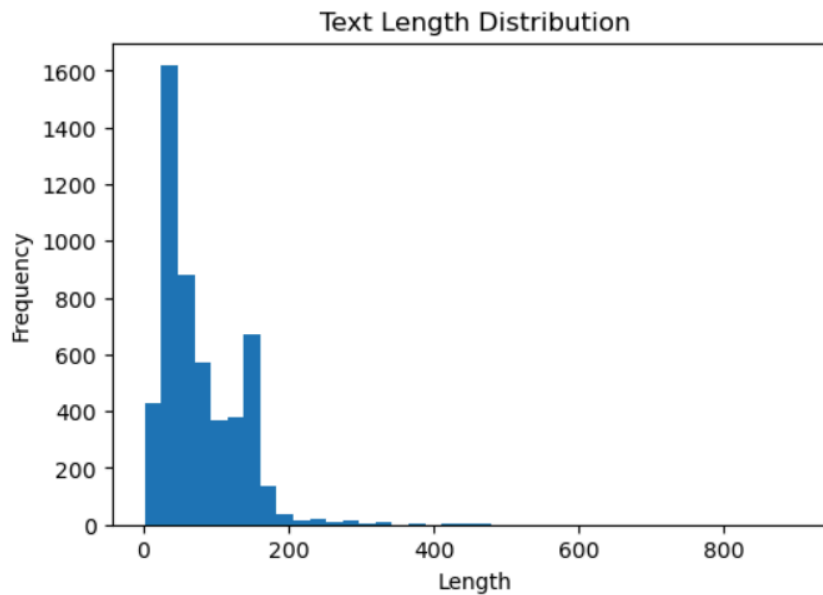
## 4.1 Class Balance

Shows the proportion of ham vs spam messages.



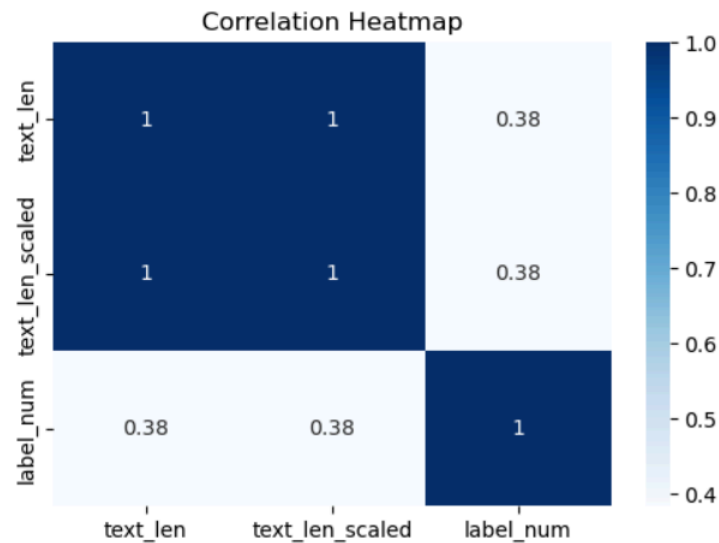
**Interpretation:** The dataset is imbalanced, with ham messages dominating.

## 4.2 Text Length Distribution



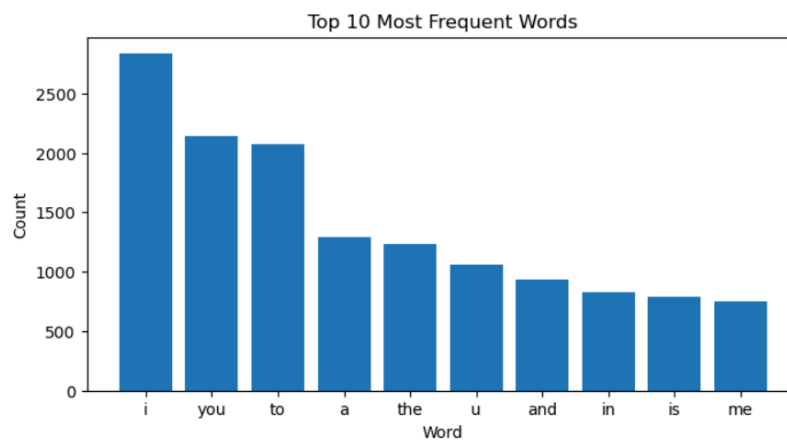
**Interpretation:** Most messages are short, matching typical SMS behavior. A long tail exists but extreme values were cleaned.

### 4.3 Numerical Feature Heatmap



**Interpretation:** Shows relationships between numeric variables. Text length is mildly correlated with the spam label.

### 4.4 Top-10 Most Frequent Tokens



**Interpretation:** Shows the most common words in the dataset. This helps understand vocabulary and potential token cleaning needs.

## 4.5 Before / After Example

```
=== BEFORE CLEANING ===
v1                                     ham
v2    Go until jurong point, crazy.. Available only ...
Name: 0, dtype: object

=== AFTER CLEANING ===
label                                     ham
text    Go until jurong point, crazy.. Available only ...
label_num                                     0
text_len                                     111
text_len_scaled                             0.549917
Name: 0, dtype: object
```

**Interpretation:** Demonstrates the effect of column selection, renaming, duplicate removal, and target encoding.

## 5. Summary:

The SMS Spam dataset was successfully loaded, inspected, and cleaned using a simple and clear pipeline. Unnecessary columns were removed, missing entries and duplicates were handled, labels were encoded numerically, and a basic text-length feature was engineered. Outliers were removed and numeric features were scaled.

The visualizations confirmed key characteristics of the dataset:

- Strong class imbalance
- Short-message distribution typical of SMS
- Clean feature correlations
- Clear vocabulary patterns

After these steps, the dataset is **clean, consistent, and ready for baseline machine-learning models**. The main consideration for modeling will be addressing class imbalance through resampling or class-weighting.